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Expert knowledge data-driven based actor–critic reinforcement learning framework to solve computationally expensive unit commitment problems with uncertain wind energy ☆

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ABSTRACT

With the expansion of power grid, unaffordable computational cost and time will pose serious challenges of time-efficient scheduling in unit commitment problem (UCP). However, existing optimization methods, i.e., mathematical programming methods and meta-heuristic algorithms, are powerless and time-consuming to handle computationally expensive UCP (CEUCP). Thus, reinforcement learning methods with strong inference and time-saving performances are motivated to solve the computationally expensive challenges in tackling CEUCPs. In this paper, a novel expert knowledge data-driven based actor–critic (AC) reinforcement learning methodology is proposed for solving CEUCPs. Specifically, in the proposed AC reinforcement learning methodology, expert knowledge, data-driven surrogate model, and improved meta-heuristic algorithm are integrated for further performance enhancement. Firstly, a novel action selection mechanism (based on the expert knowledge of thermal units characteristic) is integrated into AC to improve the efficiency of network training. Secondly, an improved extreme learning machine (ELM) data-driven surrogate model is proposed to build reward function in AC. In detail, original cost function in reward is replaced by a lightweight ELM model. Shape distance is integrated into ELM for enhancing accuracy. Finally, original marine predators algorithm (MPA) is improved for obtaining optimal dispatching decisions and rewards of AC method quickly and correctly. Original search pattern is replaced by quantum based representation for boosting convergence. The excellent performances of the proposed AC framework are verified by simulations of 10-units, 100-units, and 100-units with wind energy test systems.

1. Introduction

Operation and management of power systems are facing a critical challenge in the form of UCP, particularly with the expansion and increasing complexity of power grids [1]. Primary objective of UCP is to determine the status of units and dispatching decisions while meeting demand and minimizing costs. The significance of UCP lies in its pivotal role in guaranteeing efficient and reliable operations within a power grid. Stability, dependability, and cost-effectiveness of a power system are directly impacted by optimal commitment of thermal units [2]. Therefore, it is imperative to develop efficient methodologies for solving UCPs.

Conventional techniques employed for solving UCPs encompass priority list methods [3] and mixed-integer programming [4], etc. In recent years, meta-heuristic algorithms are employed for solving UCPs, such as teaching–learning based optimization [5], gravitational search algorithms [6], and shuffled frog-leaping algorithms [7]. However, these existing approaches are commonly suitable for solving UCPs in small-scale, low-dimension, and definite power systems. Power system is becoming progressively complex due to the expansion of modern power grid, characterized by the integration of volatile renewable energies [8]. Non-deterministic polynomial hard (NP-hard) attribute is observed within the optimization problem of modern power grid [9]. Thus, traditional UCP is transferred into CEUCPs [10]. Due to limited dispatching cycle and computing resource, both conventional methods and meta-heuristic algorithms encounter challenges when solving CEUCPs. This computationally expensive difficulty necessitates the development of more efficient approaches for tackling CEUCPs.

Data-driven methods are often employed to solve UCPs for more effective solutions. The data-driven linear power flow model was proposed in [11] which was suitable for UCPs. This method provides accurate solutions of UCPs while ensuring computational efficiency.

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Nomenclature

Parameters

F	Objective function of total cost
S	The number of scenarios
T	Total T time periods (hours)
NT	Total NT thermal units

Variables

$P_{\text{load}}(t)$	Power load at tth hour
$F_i\left(P_i^s(t)\right)$	cost of ith unit at tth hour in sth scenario
t	tth time period
i	ith thermal unit
$W_{\text{r}}^s(t)$	Wind energy at tth hour in sth scenario
$P_i^s(t)$	Power of ith unit at tth hour in sth scenario
$UR_i^s(t)/DR_i^s(t)$	Ramp-up/down rates of ith unit at tth hour in sth scenario
a_i, b_i, c_i	Parameters of cost in ith unit
$I_i(t)$	The start/stop (on/off) state (0/1 integer variable) of ith unit for tth time interval
$P_{\text{loss}}^s(t)$	Losses at tth hour in sth scenario
CH_i	Cold start time periods of ith unit
$W_{\text{re}}(t)$	Predicted wind energy at tth hour
$P_i^{\min}$	The minimum output of ith unit
$US_{\text{i}}^{\max}$	The maximum reserve contribution of ith unit
d	The percentage of maximum unit capacity
$P_i^{\max}$	The maximum output of ith unit
$US_i^s(t)$	The maximum reserve contribution of ith thermal unit at tth time period in the sth scenario
SR_t	The maximum spinning reserve at the tth hour
$\psi\left(v_w^s(t)\right)$	Wind energy output with different speed
SC_i	Startup cost of ith unit
$UR_i^{\max}$	Maximum ramp-up rate of ith unit
W_{N}	Rated wind power output
$DR_i^{\max}$	The maximum ramp-down rate of ith unit
$T_{\text{ON},i}$	The minimum start time of ith unit
$h_{\text{ON},i}(t)$	Time period that ith unit has been continuously up till tth hour
$T_{\text{OFF},i}$	The minimum down time of ith unit
$h_{\text{OFF},i}(t)$	The number of hours that ith unit has been continuously down till tth hour
HC_i/CC_i	Cost of hot/cold start in ith unit
$v_1/v_2/v_3$	Cut-in/rated/cut-out wind speed
$v_w^s(t)$	Wind speed at tth hour in sth scenario

However, the uncertain renewable energies were not considered. Furthermore, the time-consuming objective functions were not approximated by data-driven method. In [12], a data-model-driven method was proposed to solve stochastic UCPs. Gradient boosting model and deep network were employed for fitting the minimum cost. Then, model-based inverse optimization algorithm was employed for determining unit on/off status and dispatching decisions. The innovation of this study was to decouple economic dispatching from unit commitment problems while maintaining the accuracy of the solution. However, the method was systematically demonstrated on a simplified single-node system, which might need to be further validated and extended to more complex real-world power systems. In [13], a data-driven robust optimization algorithm was employed for solving UCPs considering random wind power. This hybrid method of nuclear density estimation and Wasserstein measurement was employed to obtain the dispatching decisions of UCPs. However, this method was highly dependent on data set which could lead to inaccurate dispatching decisions. In [14], a data-driven variable reduction method was proposed to effectively solve the transmission constrained UCPs in large-scale systems. However, the efficiency of calculation was not sufficient and the time of calculation could be further reduced. To sum up, for solving CEUCPs, more effective and time-savingly data-driven methods deserve further study to obtain dispatching decisions.

Reinforcement learning (RL) and machine learning (ML) are motivated to solve the aforementioned challenges in tackling CEUCPs. Formulation of sequential unit status in RL and ML can be achieved efficiently, even in the absence of an accurate environment model. To gain a more comprehensive understanding and accurately depict the characteristics of the environment, ML methodologies can be incorporated to facilitate more precise decisions in CEUCPs [15]. In [16], a ML approach employing recurrent neural networks was utilized to efficiently solve the UCPs involving stochastic wind energy. However, the study only focused on six thermal units and one wind farm, thus limiting its applicability to large-scale power grids. The resolution of UCP was facilitated by employing a trained ML model in [17], specifically based on the K-nearest neighbor algorithm. The security of the random dispatching was demonstrated. However, the aspect of cost-effectiveness was not taken into consideration. In [18–20], various RL approaches were proposed to solve the challenges for solving UCPs. These approaches effectively determined accurate unit status and provided optimal dispatching solutions. However, the maximum number of units was only 30. Due to the interconnection of modern power systems, the scale of units and variables are significantly increased, resulting in computationally expensive features. The aforementioned RL methods may become less effective in solving complex CEUCPs. Therefore, there is a pressing necessity for a RL method that can effectively handle CEUCPs with high-dimension units and variables.

In detail, the challenge of effectively and efficiently solving CEUCPs by RL methods can be attributed to three key factors: **(i)** High-dimension 0/1 variables lead to "curse of dimensionality" within determining and optimizing the states of units. The dimensions of actions in RL applications are extremely high that traversing them becomes challenging; **(ii)** Existing approaches often fail to ensure convergence and robustness across different time periods in CEUCPs, leading to significant economic losses. In other words, the rewards in RL applications may be imprecise which can lead to non-convergence of RL algorithm; **(iii)** Previous studies often fail to effectively solve CEUCPs in a short time. Consequently, their applicabilities in real-world electric power generation scenarios are limited.

To solve the action traversal problem in RL for solving CEUCPs, expert knowledge based action selection mechanisms can be employed. In [21], controller and guidance low-based expert knowledge were combined to restrict action space. However, the guidance low-based expert knowledge primarily solved continuous actions and provided limited assistance in generating discrete actions of CEUCPs. The expert knowledge of extreme weather events is employed by Ref. [22] to solve the issue of handling discrete actions in power system redispatching. However, this investigation was limited to a small-scale grid. The structural-based expert knowledge was proposed in [23] to enhance efficacy of action choice. However, a low level of convergence was still remained in RL. To sum up, effective expert knowledge based action selection mechanisms is deserved to be studied in RL application for solving CEUCPs.

To improve accuracy of dispatching cost across different time periods in CEUCPs, stable meta-heuristic algorithm can be employed [24]. Economic dispatch (ED) tasks exhibit variations across different time periods, which present significant challenges to convergence speed, convergence time, and robustness of meta-heuristic algorithms in achieving optimal dispatching decisions. Total cost of CEUCP is significantly impacted by convergence and robustness performance of a meta-heuristic algorithm. Therefore, an appropriate meta-heuristic algorithm is crucial in solving CEUCPs. In [25], a novel binary fish migration optimization algorithm was proposed to solve CEUCPs. It obtained excellent performance in several benchmark systems. However, its effectiveness was limited when applied to the IEEE 100 units test system due to certain deficiencies in solving CEUCPs with large-scale units. Ref. [26] introduced an enhanced genetic algorithm to solve the challenges of CEUCPs. Although this approach successfully reduced hourly cost, the approximation to unit status resulted in suboptimal and imprecise solutions. In [27], a coyote optimization algorithm was proposed to solve CEUCPs. However, it failed to incorporate volatility of renewable energy and exhibited limited robustness for hourly dispatching. In conclusion, meta-heuristic algorithms hold great potential for solving CEUCP when the robustness and convergence are improved.

To mitigate the time-consuming limitation of optimization in CEUCP [28], surrogate model can be employed [29]. The evaluation time of objective function in CEUCP is considerably lengthy, which contradicts the requirement for expedited scheduling. The determination of unit status requires a significant amount of time for evaluations of objective functions within CEUCPs. In recent years, surrogate-assisted techniques obtain widespread recognition due to time-saving ability to significantly reduce computational time for large-scale optimization problems. In [30], a neural network surrogate model was proposed to substitute fuel cost function in ED problem resulting in a significant reduction of dispatching time. However, obtained dispatching decisions exhibited slight inadequacy from an economic perspective. Support vector regression surrogate models were proposed to assist multi-area economic/emission dispatch optimization in [31]. Much-reduced computing time and more accuracy solutions were obtained by this surrogate-based method. However, it did not take into account 0/1 integer variables. This deficiency made it unsuitable for solving CEUCPs. In addition, Refs. [30,31] failed to account for uncertainty associated with renewable energy sources, which were essential for meeting clean and low-carbon requirements of modern power systems [32]. In conclusion, it is urgently needed to build surrogate models for efficient evaluation of objective functions involving a large number of 0/1 integer variables and uncertain renewable energy in the context of CEUCPs.

Total three main gaps are summarized by the above analyses:

(1) The application of RL methods with exceptional convergence performance and speed to CEUCPs is explored barely. Though expert knowledge based action selection mechanisms are employed by Ref. [21,23] to accelerate convergence speed in RL, these two methods are powerless in obtaining robust and economic dispatching decisions in CEUCPs. Ref. [22] solved the action traversal problem in CEUCPs. However, the limited low-dimension application scenario is not suitable to high-dimension power system.
(2) The robustness and convergence of most meta-heuristic algorithms can hardly be guaranteed for every dispatching time period in CEUCPs. In [5,6], and [7], the ability of jumping out of locally optimal solution is insufficient. This drawback leads to uneconomical dispatching decisions. The robustness of meta-heuristic algorithms proposed in Ref. [25,26], and [27] is insufficient for solving CEUCPs, which may bring infeasible optimal solutions.
(3) Emerging computationally expensive property in UCPs is barely handled by previous studies. Data-driven surrogate-assisted methods in [30,31] are powerless to solve CEUCPs with a large number of 0/1 integer variables. Furthermore, the significance of renewable energy is ignored by these researches. The time-efficient method for solving computationally expensive dilemma in CEUCPs is needed urgently.

To fill the proposed three research gaps, unit status in CEUCPs is determined through applying a RL framework in this paper. To solve the action traversal problem in RL, expert knowledge based action selection mechanism which is suitable for solving CEUCPs should be studied. To enhance computational efficiency, surrogate-assisted methodologies can be employed to solve ED problems within CEUCPs, by integrating a meta-heuristic algorithm with a surrogate model. In this paper, an improved deep reinforcement learning framework is presented for solving CEUCPs. Initially, a refined AC reinforcement learning framework is introduced to determine operational status of units. The novel expert knowledge based action selection mechanism draws inspiration from the characteristics of thermal units. Subsequently, improvements are made to MPA for identifying dispatching decisions. Lastly, original objective function in CEUCP is reconstructed by an improved and time-saving ELM surrogate model. Major outcomes are listed as follows:

(1) A novel expert knowledge data-driven based AC reinforcement learning framework is proposed for solving CEUCPs to determine hourly status of units. To overcome the difficulty of non-convergence caused by complex and exponential actions, a novel action selection mechanism based on the expert knowledge of thermal unit characteristic is designed. Unlike a data science perspective, this artificial intelligence method is specifically tailored for unit commitment problems and assisted by expert knowledge of industrial production experience. The training speed and model performance are enhanced through the utilization of this innovative action selection mechanism. By applying the proposed AC reinforcement learning framework, more economically optimal commitments can be obtained within seconds. CEUCPs containing high-dimension 0/1 variables are solved effectively.
(2) Original MPA is enhanced with a pioneering quantum uncertainty-based search approach to meet the precision requirements of dispatching decisions in CEUCPs. Additionally, it is utilized for obtaining the rewards of AC. In order to achieve accurate and robust dispatching decisions in CEUCPs, a pioneering quantum uncertainty-based search approach is integrated into MPA to replace original local search process. The improved algorithm is named quantum-marine predators algorithm (Q-MPA). This enhancement not only improves the robustness of Q-MPA but also enhances its convergence by leveraging a quantum uncertainty-based search pattern.
(3) A novel data-driven optimization method based on ELM is employed to solve CEUCPs. An enhanced data-driven surrogate model based on ELM is proposed as a replacement for the original cost function in CEUCPs. ELM surrogate model is trained using historical data pertaining to unit status (comprising 0/1 integer variables) and dispatching decisions (continuous variables). To enhance the accuracy of surrogate model, shape distance is incorporated into the original ELM as an additional convergence criterion. Accurate construction of ELM surrogate model (encompassing abundant 0/1 variables) is achieved through this approach. Consequently, both evaluation time for the cost function and optimization time for solving CEUCPs are significantly reduced.

The remainder of this paper is organized as follows. Section 2 describes the CEUCPs in the expanded power grid. Section 3 proposes AC reinforcement learning, Q-MPA, and ELM-based surrogate models. Section 4 provides simulation results of the 10 units test systems and 100 units test systems with or without wind energy of CEUCPs. Section 5 summarizes the paper and suggests future research directions.

2. Objectives and constraints of CEUCPs

This section describes the CEUCPs with uncertain wind energy. Many equation and inequality constraints are contained in CEUCPs with uncertain wind energy. CEUCPs are formulated as following equations and inequations.

2.1. Objective function

Objective function of CEUCP

In CEUCP, objective is minimizing total cost of units, including fuel and start-up cost.

$$F = \sum_{s=1}^{\mathrm{S}} \sum_{t=1}^{\mathrm{T}} \sum_{i=1}^{\mathrm{NT}} \left[I_i(t) \times F_i \left(P_i^s(t) \right) + I_i(t) \times \left(1 - I_i(t-1) \times SC_i \right] \tag{1}$$

Objective function of ED in CEUCP

The objective function of fuel cost $F_i\left(P_i^s(t)\right)$ can be described as a quadratic function.

$$F_i\left(P_i^s(t)\right) = a_i \left(P_i^s(t)\right)^2 + b_i P_i^s(t) + c_i \tag{2}$$

2.2. Constraints

Power balance limit

Power balance limit represents that a balance between total outputs of all units and the sum value of load demand and transmission loss should be strike.

$$\sum_{i=1}^{\mathrm{NT}} I_i(t) \times P_i^s(t) + W^s(t) = P_{\mathrm{load}}(t) + P_{\mathrm{loss}}^s(t) \tag{3}$$

Generation limit

Generation limit represents that the unit output must belong to the maximum and minimum unit output.

$$P_i^{\min}(t) \times I_i(t) \le P_i^s(t) \le P_i^{\max}(t) \times I_i(t) \tag{4}$$

Spinning reserve constraints

Spinning reserve constraints represent that total remaining capacity of all online units should be larger than the default value for emergencies.

$$\begin{cases} US_{\mathrm{i}}^{\max} = d \times P_i^{\max} \\ US_i^s(t) = \min\{US_i^{\max}, P_i^{\max} - P_i^s(t)\} \\ \sum_{i=1}^{\mathrm{NT}} I_i(t) \times US_i^s(t) \ge SR_t \end{cases} \tag{5}$$

Ramping up/down capacity constraints

Ramping up/down capacity constraints represent that sudden change in power generation cannot occur for safe and stable operation.

$$\begin{cases} UR_i^s(t) = \min\{UR_i^{\max}, P_i^{\max} - P_i^s(t)\} \\ DR_i^s(t) = \min\{DR_i^{\max}, P_i^s(t) - P_i^{\min}\} \end{cases} \tag{6}$$

The minimum start/stop hours constraints

The minimum start/stop hours constraints represent that the units cannot be turned on and off frequently in a short period of time.

$$\begin{cases} T_{\mathrm{ON},i} \le h_{\mathrm{ON},i}(t) \\ T_{\mathrm{OFF},i} \le h_{\mathrm{OFF},i}(t) \end{cases} \tag{7}$$

Startup cost

Startup cost of thermal power units is closely relevant to unit boiler temperature. Startup cost will increase if boiler keeps in cold state for a long time.

$$SC_i = \begin{cases} HC_i, & T_{\mathrm{OFF},i} \le h_{\mathrm{OFF},i} \le T_{\mathrm{OFF},i} + CH_i \\ CC_i, & h_{\mathrm{OFF},i} > T_{\mathrm{OFF},i} + CH_i \end{cases} \tag{8}$$

Power output limit on wind energy system

This constraint represents that output function of wind farm is segmented according to the cut-in, rated, cut-out wind speeds.

$$W_{\mathrm{r}}^s(t) = \begin{cases} 0, & v_w^s(t) \le v_1 \\ \psi\left(v_w^s(t)\right), & v_1 \le v_w^s(t) \le v_2 \\ W_{\mathrm{N}}, & v_2 \le v_w^s(t) \le v_3 \\ 0, & v_w^s(t) \ge v_3 \end{cases} \tag{9}$$

2.3. Monte Carlo method to model wind energy

The continuous wind scenario is hard to describe. Monte Carlo simulation can be employed to generate the scenarios between the given bands (75% to 125%). Then, the reduction scenarios are generated by fast pre-generation elimination method based on probabilistic distance. In this paper, the predicted wind energy [33] is shown as Fig. 1(a), and the generated 2000 scenarios are shown in Fig. 1(b). The Monte Carlo simulation can be described as Algorithm 1. Then, total twenty typical scenarios shown in Fig. 2(a) are generated by fast pre-generation elimination method based on probabilistic distance, and respective probabilities are shown as Fig. 2(b). Reduction method is described as Algorithm 2.

Algorithm 1: Pseudo of Monte Carlo simulation

Input: $W_{\mathrm{re}}(1), W_{\mathrm{re}}(2), \ldots, W_{\mathrm{re}}(t), \ldots, W_{\mathrm{re}}(\mathrm{T})$
Output: Total S scenarios
for s=1:S **do**
  for t=1:T **do**
    Generate $W^s(t)$ randomly by uniform distribution within $[0.75W_{\mathrm{re}}(t), 1.25W_{\mathrm{re}}(t)]$
    $W^s(t)$ is recorded as the actual wind generation at tth hour in sth scenario
  end
end

As shown in Algorithm 1 and Fig. 1(b), Monte Carlo simulation method is employed for constructing uncertain scenarios of wind energy. Predicted wind energy in a day can be described as $W_{\mathrm{re}}(1)$, $W_{\mathrm{re}}(2), \ldots, W_{\mathrm{re}}(t), \ldots, W_{\mathrm{re}}(\mathrm{T})$, where $W_{\mathrm{re}}(t)$ is predicted wind energy at tth time period, T is the total time periods (in this paper, T is equal to 24 and represents total 24 h). For the sth scenario, the wind energy output $W^s(t)$ in tth time period is generated randomly by uniform distribution within $[0.75W_{\mathrm{re}}(t), 1.25W_{\mathrm{re}}(t)]$. The wind energy output of sth scenario can be generated by cycling the above sampling process 24 times (total 24 h). Assuming that a total of S uncertainty scenarios are required, the above loop process can be repeated S times. Then, total S wind energy uncertainty scenarios can be obtained.

The uncertainty scenarios generated according to the above scheme are not necessarily representative. Thus, scenario reduction method is needed to reduce unnecessary computational overhead as shown in Algorithm 2 and Fig. 2(a). Specifically, assuming that a total of $\mathrm{S_r}$ uncertainty scenarios are chosen in S wind energy uncertainty scenarios. Firstly, all Euclidean distance $d_{s,ss}$ are calculated between scenario s and scenario ss ($s \neq ss$). Secondly, the scenario $s_{\min}$ which has the minimum sum of distances from the remaining scenarios is chosen. Thirdly, the scenario $s_{\min}$ is replaced by scenario $s_{\mathrm{min,n}}$, where scenario $s_{\mathrm{min,n}}$ has the minimum distance to $s_{\min}$. Finally, original probability of scenario $s_{\min}$ is added to $s_{\mathrm{min,n}}$ as shown in Fig. 2(b), and an uncertainty scenario can be reduced. By performing the above four steps, the number of wind energy uncertainty scenarios is reduced to $\mathrm{S_r}$.

Algorithm 2: Pseudo of reduction method

Input: Total S scenarios
Output: $\mathrm{S_r}$ typical scenarios after reduction
while Number of remaining scenarios $\le \mathrm{S_r}$ **do**
  Calculates all Euclidean distance $d_{s,ss}$ between scenario s and scenario ss ($s \neq ss$)
  Select the scenario $s_{\min}$ which has the minimum sum of distances from the remaining scenarios
  Replace the scenario $s_{\min}$ by scenario $s_{\mathrm{min,n}}$, scenario $s_{\mathrm{min,n}}$ has the minimum distance to $s_{\min}$
  Add the probability of scenario $s_{\min}$ to $s_{\mathrm{min,n}}$
end

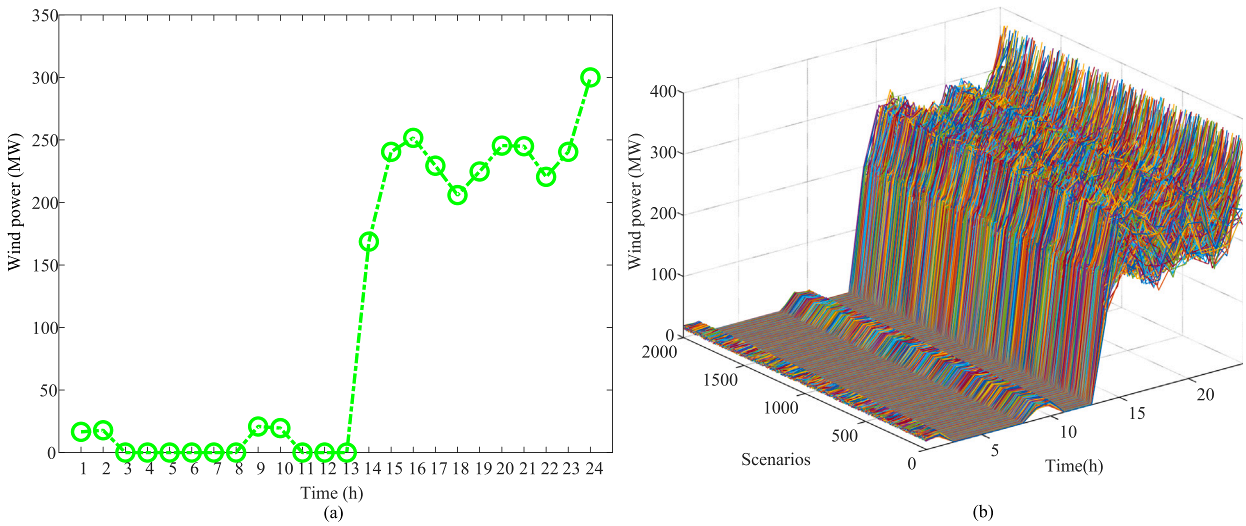

Fig. 1. (a) Predicted wind power in this paper. (b) The uncertain wind energy scenarios in this paper.

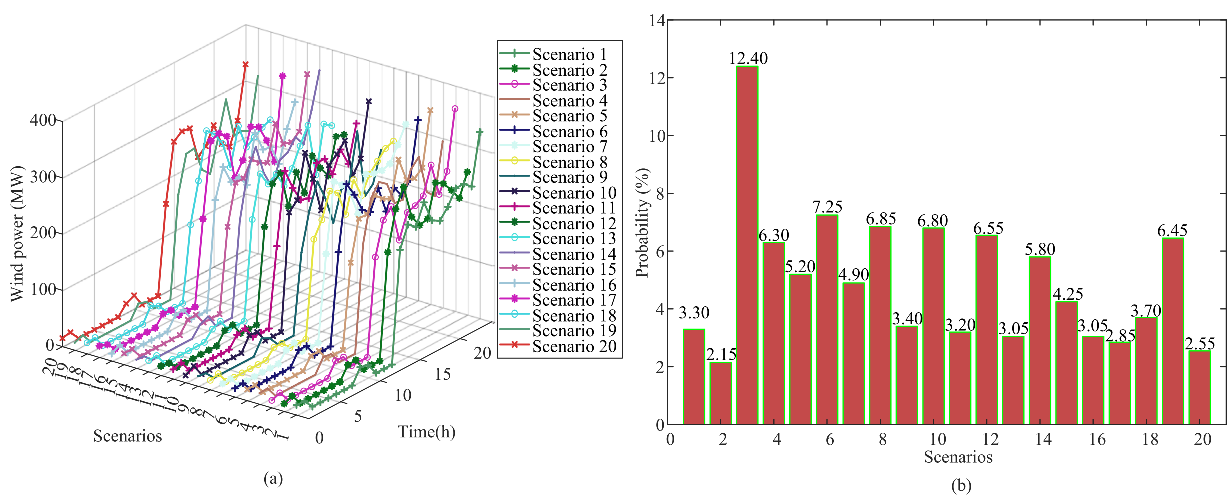

Fig. 2. (a) Typical scenarios after reduction in this paper. (b) The probabilities of typical scenarios.

3. Descriptions of proposed method

In this section, original CEUCP with uncertain wind energy is presented through Markov decision process (MDP). Then, CEUCP is handled through an improved AC method with a new action selection mechanism. The reward value in AC reinforcement learning is computed by Q-MPA. To further reduce time, original cost function in CEUCP is replaced by an improved ELM surrogate model.

3.1. *The expert knowledge data-driven based AC framework to solve CEUCPs*

Generally, CEUCPs are treated as discrete-time stochastic programming processes. According to the accumulated rewards along discrete period, the decision series of CEUCPs can be determined. A partial observable MDP is described as a five-tuple ($\mathcal{S}$, $\mathcal{A}$, $\mathcal{S}'$, $\mathcal{P}$, $\mathcal{R}$), where $\mathcal{S}$ denote the sets of states, $\mathcal{A}$ denote the sets of actions, $\mathcal{S}'$ denote the sets of next states, $\mathcal{P}$ is a set of conditional transition probabilities between states, and $\mathcal{R}$ is the reward function. Furthermore, a CEUCP can be described as a partial observable MDP, and model-free method is used to train the agent. The CEUCP agent can be put to the real-world environment and make online judgments effectively after being well trained [34].

(a) *The MDP description of CEUCPs with wind energy*

- State: Observable state $\mathcal{S}(t)$ includes up/down time $[h_{\mathrm{ON/OFF},1}(t), \ldots, h_{\mathrm{ON/OFF,NT}}(t)]$ for NT thermal units.
- Action: The action in the proposed method is corresponding to turning on or off the NT thermal units. Specifically, action includes a series of 0/1 discrete variables $[I_1(t+1), \ldots, I_{\mathrm{NT}}(t+1)]$.
- Reward: The reward is the result of dividing cost by current power demand at tth hour. The cost is the sum of operating, cold start-up, and hot start-up costs at the tth hour. Further, to obtain operating cost quickly, Q-MPA is proposed to execute ED with determined unit status. The description of Q-MPA is shown in Section 3.2. To further reduce execution time, original cost functions is replaced by an ELM surrogate model. Description of the ELM is shown in Section 3.3.

(b) *Deep RL algorithm to solve CEUCPs*

Actor–critic networks are used to obtain value and behavior estimations.

- Critic network: Value estimation is handled by the critic network. In order to assist agents in taking the action in the next step, critic network calculates potential action values based on the current state. Concretely, convolutional neural network (CNN) architecture is to estimate Q-value pairs ($Q(state, action)$), where

$Q(state, action)$ is the possible reward for selecting an action in current state.

- Actor network: The structure of actor network is CNN. In contrast to critic network, actor network predicts behavior distribution when an agent enters a new state. Then, based on the feedback provided by critic network, actor network takes required action.
- Update rule: ① Update rule of critic network: In the critic network, q value (output value) is shown as:

$$q = Q(state, action|\phi) \tag{10}$$

where q is evaluation value, $Q(state, action|\phi)$ is the output when input $action$, and ϕ represents parameters and structure of critic network. Then, the evaluation of state–action pair $(state, action)$ can be described as:

$$Q(state, action) = reward + \gamma^{\star} Q\left(state', action'\right) \tag{11}$$

where $Q\left(state', action'\right)$ is the Q value of next state–action pair $\left(state', action'\right)$, $reward$ is the true reward, and $\gamma^{\star}$ is the discount factor. Then, ϕ is updated by making the output of $Q(state, action|\phi)$ close to $reward + \gamma^{\star} Q\left(state', action'\right)$. The difference value is temporal difference (TD) error:

$$\text{TD\_error} = Q(state, action) - Q(state, action|\phi) \tag{12}$$

② Update rule of actor network: In actor network, the output $action$ is shown as (13).

$$action = \pi(state|\theta) \tag{13}$$

where $state$ is the input state, θ is the parameters and structure of the actor network, and $\pi(state|\theta)$ represents the policy. In order to update actor network, θ should be adjusted to obtain a better $\pi(state|\theta)$ when a particular $state$ appears. The direction of adjustment is to obtain a larger $Q(state, \pi(state, \theta)|\phi)$, where the $Q(state, \pi(state, \theta)|\phi)$ is the output of critic network when input the $\pi(state, \theta)$. When critic network parameter ϕ is constant and the $state$ is fixed, Q is a composite function of θ, which can be obtained by (14). J is gradient, and θ is updated up along J.

$$\mathrm{J} = \frac{\delta \mathrm{Q}}{\delta \theta} = \frac{\delta \mathrm{Q}(\mathrm{state}, \mathrm{action})}{\delta \pi(\mathrm{state} \mid \theta)} \times \frac{\delta \pi(\mathrm{state} \mid \theta)}{\delta \theta} \tag{14}$$

(c) *Expert knowledge based action selection mechanism in AC reinforcement learning*

In AC reinforcement learning, the next action of current state is chosen according to the best evaluation value of critic network. It is an inefficient action selection mechanism [35]. To overcome this drawback, an expert knowledge based action selection mechanism is proposed to improve efficiency of action selection. In CEUCPs, the action is represented by a series of 0/1 discrete variable. When a state occurs, the on/off states of one or two thermal unit can be changed while meeting power load and constraints. The difference between original action and the new actions can be measured by Hamming distance. Based on the above analyses, this mechanism is proposed to accelerate convergence. Pseudo codes and framework of AC are shown as Algorithm 3 and Fig. 3, respectively. In [19], a deep AC framework is also proposed to solve UCPs with uncertain wind energy. Compared with this method, the proposed AC method include four distinct differences: ① Compared with the epsilon-greedy method for sampling, a power system expert knowledge based sampling method is proposed, which is more suitable for solving UCPs.② Compared with the gradient-based method to obtain rewards, meta-heuristic algorithm is employed to obtain more accurate rewards in this paper. ③ Compared with the approximatively numerical reward function, more accurate and time-saving ELM surrogate model is proposed to replace reward function in this paper. ④ More important, compared with small-scale UCPs, the proposed method in this paper focus on the large-scale and high-dimension CEUCPs.

Algorithm 3: AC method with proposed expert knowledge based action selection mechanism

Input: Original θ and ϕ
Output: Trained θ and ϕ

```
for i = 1 : Maximum iterations do
    Use policy π(state, θ) to obtain an action
    Use proposed expert knowledge based action selection
    mechanism to obtain actions
    Calculate evaluation values
    Store best solution
    if Solutions in replay buffer achieves its upper limit then
        Update θ and φ and clear stored solutions
    end
end
```

3.2. Q-MPA to obtain dispatching decisions

In the training of AC reinforcement learning, the reward depends on the total cost at tth time period. Thus, a robust and precise meta-heuristic algorithm is essential for obtaining dispatching decisions. MPA is an effective and efficient evolutionary algorithm for optimization task. MPA obtains excellent features, such as no-derivative, no-parameter, flexibility, stability, and simplicity [36]. The no-derivative and no-parameter features represent that MPA can handle most optimization problems (include CEUCPs). The flexibility represents that the MPA can adjust the search scheme adaptively for solving CEUCPs [37]. It means that more economical dispatching decisions of CEUCPs may be obtained. Stability indicator is especially important to solve CEUCPs [38]. Power system operation depends on the stable optimal solution. Thus, MPA obtains broad prospects for solving CEUCPs in modern power systems.

However, for CEUCPs, the convergence and robustness of MPA are still insufficient [39]. To overcome these drawbacks, original local search process in MPA is replace by a quantum uncertainty-based search approach which named Q-MPA. The Q-MPA is used to obtain rewards of AC reinforcement learning and dispatching decisions for CEUCPs.

Q-MPA simulates the foraging behavior of marine predators. The optimization is executed by searching, tracking, and attacking prey. Q-MPA randomly generates individuals in the search space through (15):

$$X^{\mathrm{u}} = \mathrm{P}^{\min} + \mathrm{rand}\ (\cdot) \times (\mathrm{P}^{\max} - \mathrm{P}^{\min}) \tag{15}$$

where $\mathrm{P}^{\min}$ and $\mathrm{P}^{\max}$ are lower and upper limits (considering the ramping up/down capacity constraints) for thermal units, $\mathrm{rand}\ (\cdot)$ is a uniformly random vector within [0, 1], and X^{u} is an individual. If the on/off state of a unit is off, the value of it is set as 0. Population is shown as $Prey$ matrix:

$$Prey = \begin{bmatrix} X^{\mathrm{u}}_{1,1} & X^{\mathrm{u}}_{1,2} & \cdots & X^{\mathrm{u}}_{1,\mathrm{NT}} \\ X^{\mathrm{u}}_{2,1} & X^{\mathrm{u}}_{2,2} & \cdots & X^{\mathrm{u}}_{2,\mathrm{NT}} \\ \vdots & \vdots & \vdots & \vdots \\ X^{\mathrm{u}}_{n,1} & X^{\mathrm{u}}_{n,2} & \cdots & X^{\mathrm{u}}_{n,\mathrm{NT}} \end{bmatrix}_{n \times \mathrm{NT}} \tag{16}$$

where n is the number of individuals and NT is dimension (in CEUCPs, NT is the number of thermal units).

The individual whose total cost is the minimum value in current iteration is chosen as elite individual (X^{l}) . This individual is replicated n times to construct the $Elite$ matrix:

$$Elite = \begin{bmatrix} X^{\mathrm{l}}_{1,1} & X^{\mathrm{l}}_{1,2} & \cdots & X^{\mathrm{l}}_{1,\mathrm{NT}} \\ X^{\mathrm{l}}_{2,1} & X^{\mathrm{l}}_{2,2} & \cdots & X^{\mathrm{l}}_{2,\mathrm{NT}} \\ \vdots & \vdots & \vdots & \vdots \\ X^{\mathrm{l}}_{n,1} & X^{\mathrm{l}}_{n,2} & \cdots & X^{\mathrm{l}}_{n,\mathrm{NT}} \end{bmatrix}_{n \times \mathrm{NT}} \tag{17}$$

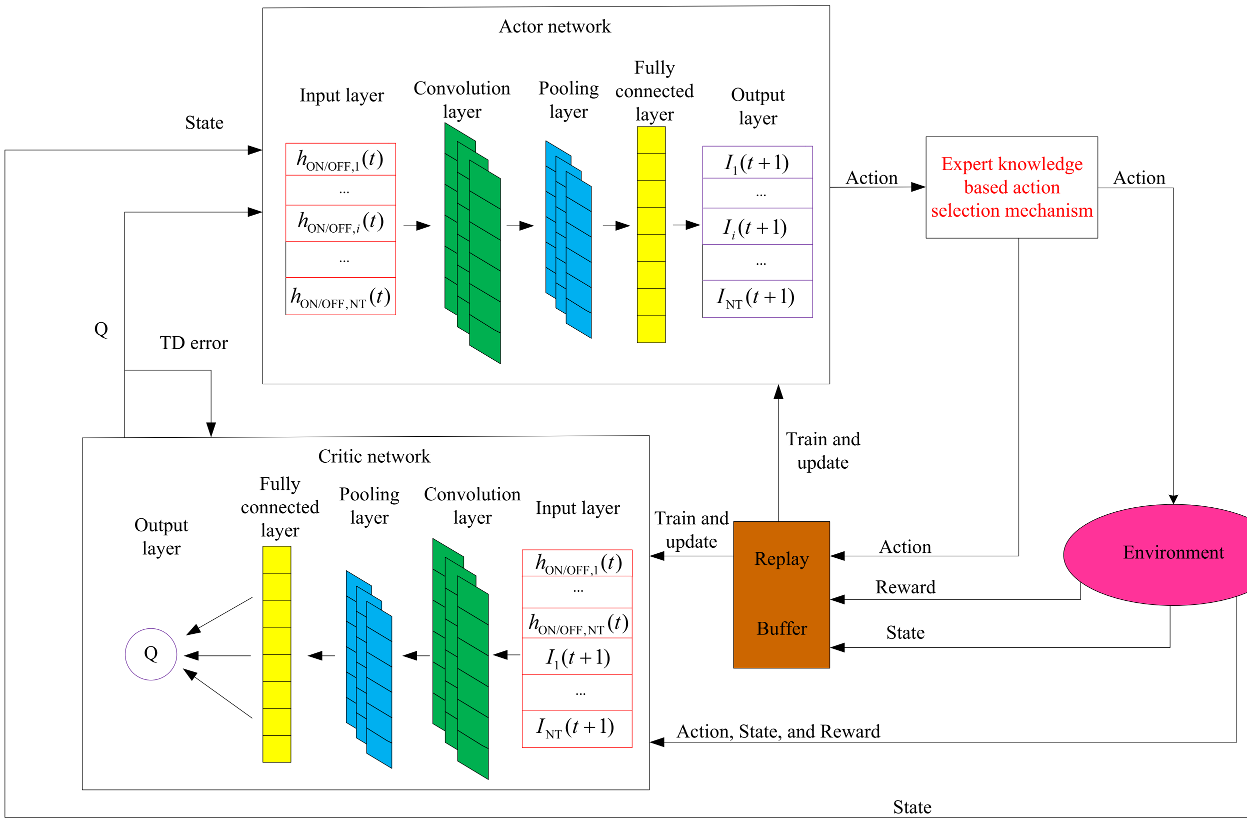

Fig. 3. Schematic diagram of AC reinforcement learning for CEUCPs.

Total three stages are contained in optimization process: high, unit, and low velocity ratio stages. Firstly, high velocity ratio stage means that iteration is less than a third of the maximum iterations:

$$\begin{cases} \overrightarrow{\mathrm{Step}}_i = \vec{\mathrm{R}}_{\mathrm{A}} \otimes (\overrightarrow{\mathrm{Elite}}_i - \vec{\mathrm{R}}_{\mathrm{A}} \otimes \overrightarrow{\mathrm{Prey}}_i) \\ \overrightarrow{\mathrm{Prey}}_i = \overrightarrow{\mathrm{Prey}}_i + \mathrm{P_R} \times \vec{\mathrm{R}} \otimes \overrightarrow{\mathrm{Step}}_i \end{cases} \tag{18}$$

where $\vec{\mathrm{R}}_{\mathrm{A}}$ is a randomly normal vector, $\otimes$ represents entry-wise multiplications, $\mathrm{P_R}$ is mixed as 0.5, $\vec{\mathrm{R}}$ is a randomly uniform vector within [0, 1], $\overrightarrow{\mathrm{Step}}_i$ is step size of ith individual, $\overrightarrow{\mathrm{Elite}}_i$ and $\overrightarrow{\mathrm{Prey}}_i$ are ith individuals in *Elite* and *Prey* metrics, respectively.

Secondly, unit velocity ratio stage means that iteration is between one third and two third of the maximum iterations. Update equations in the top half of *Prey* are:

$$\begin{cases} \overrightarrow{\mathrm{Step}}_i = \vec{\mathrm{R}}_{\mathrm{Le}} \otimes (\overrightarrow{\mathrm{Elite}}_i - \vec{\mathrm{R}}_{\mathrm{Le}} \otimes \overrightarrow{\mathrm{Prey}}_i) \\ \overrightarrow{\mathrm{Prey}}_i = \overrightarrow{\mathrm{Prey}}_i + \mathrm{P_R} \times \vec{\mathrm{R}} \otimes \overrightarrow{\mathrm{Step}}_i \end{cases} \tag{19}$$

where the $\vec{\mathrm{R}}_{\mathrm{Le}}$ is a random vector based on Levy distribution. In Q-MPA, update equation in the last half of *Prey* is replaced by a quantum based representation to enhance convergence and robustness:

$$\begin{cases} \overrightarrow{\mathrm{Step}}_i = \vec{\mathrm{R}}_{\mathrm{A}} \otimes (\vec{\mathrm{R}}_{\mathrm{A}} \otimes \overrightarrow{\mathrm{Elite}}_i - \overrightarrow{\mathrm{Prey}}_i) \\ \overrightarrow{\mathrm{Prey}}_i = \overrightarrow{\mathrm{Prey}}_i + 0.2 \times \vec{\mathrm{R}} \otimes \overrightarrow{\mathrm{Step}}_i \times \log(1/u) \end{cases} \tag{20}$$

where u is set as a random number within [0, 1]. Finally, the low velocity ratio stage means that iteration is larger than two third of the maximum iterations. In Q-MPA, the update equation is also replaced by a quantum based representation:

$$\begin{cases} \overrightarrow{\mathrm{Stepsize}}_i = \vec{\mathrm{R}}_{\mathrm{L}} \otimes (\vec{\mathrm{R}}_{\mathrm{L}} \otimes \overrightarrow{\mathrm{Elite}}_i - \overrightarrow{\mathrm{Prey}}_i) \\ \overrightarrow{\mathrm{Prey}}_i = \overrightarrow{\mathrm{Elite}}_i + 0.2 \times \vec{\mathrm{R}} \otimes \overrightarrow{\mathrm{Stepsize}}_i \times \log(1/u) \end{cases} \tag{21}$$

where $\mathrm{R_L}$ represents another random vector based on Levy distribution. Other definitions of variables are the same as the above-mentioned texts.

At the end of iteration, the objective value of X^l is the minimum total cost. Then, the reward can be calculated through dividing the minimum total cost by the power demand at current time period. To illustrate the effectiveness of Q-MPA, unimodal test functions (TF1–TF7) are utilized to test convergence. High-dimension multimodal test functions (TF8–TF13) are utilized to test search ability of Q-MPA. TF1–TF8 are employed to search the minimum value. The average (Ave) best result represents the average value of 30 experiments searching for the minimum value. Ave is employed for evaluating the accuracy of algorithm. Smaller Ave represents the better search ability. The best standard deviation (Std) is employed for evaluating error size for each result in an independent experiment. Std is employed for testing the stability of algorithm. Smaller Std represents the better robustness. In CEUCPs, on one hand, the better Ave indicator represents that more economical dispatching decisions can be obtained by the algorithm. On the other hand, the better Std indicator represents that more robust dispatching decisions can be obtained by the algorithm. Thus, in this paper, both Ave and Std indicators are employed for evaluating algorithm performance. Then, the original MPA, particle swarm optimization (PSO), an improved differential evolution (SHADE), and salp swarm algorithm (SSA) are used to compare. For all algorithms, the size of populations is 500. The number of iterations is 500. The average (Ave) best results shown in Table 1 are all obtained by Q-MPA. It illustrates that convergence of Q-MPA is excellent. The best standard deviations (Std) are all obtained by Q-MPA. It illustrates that the robustness of Q-MPA is better than other algorithms. To sum up, Q-MPA is suitable for obtaining dispatching decisions in CEUCPs.

In [40], spherical search algorithm, opposition-based learning, and differential evolution method are integrated into MPA for solving ED problem with load, solar, and wind uncertainties. Different from this improved MPA, the proposed Q-MPA in this paper enhance the search ability. Further, Q-MPA is suitable to solve high-dimension optimization problems. The improved MPA is only tested in small-scale economic dispatch problems. In [41], a random possibility-based MPA is

Table 1
Comparative results and parameter settings.

Function	Indicator	Q-MPA	MPA	PSO	SSA	SHADE
TF1	Ave	**1.64E−92**	3.27E−21	4.09E−02	3.70E−03	1.08E−08
	Std	**2.19E−92**	4.61E−21	4.16E−02	9.74E−03	1.17E−08
TF2	Ave	**1.52E−50**	1.57E−12	6.59E−02	5.05E+00	1.23E−01
	Std	**4.23E−50**	1.42E−12	8.64E−02	2.01E+00	1.85E−01
TF3	Ave	**1.05E−49**	8.64E−02	4.24E+03	4.34E+03	2.65E+02
	Std	**3.41E−49**	1.44E−01	1.22E+03	2.14E+03	1.27E+02
TF4	Ave	**2.46E−38**	2.60E−08	9.34E+00	1.51E+01	1.64E+00
	Std	**5.30E−38**	9.25E−09	1.01E+00	3.20E+00	5.35E−01
TF5	Ave	**4.58E+01**	4.60E+01	3.10E+02	4.34E+02	6.10E+01
	Std	**2.36E−01**	4.22E−01	4.31E+01	4.58E+01	3.55E+01
TF6	Ave	**2.78E−01**	3.98E−01	5.89E−02	2.10E−03	8.32E−09
	Std	**9.24E−02**	1.91E−01	1.22E−01	3.00E−03	1.00E−08
TF7	Ave	**5.93E−04**	1.80E−03	6.65E−02	2.81E−01	2.94E−02
	Std	**3.46E−04**	1.00E−03	1.23E−02	9.11E−02	1.00E−02
TF8	Ave	**−1.49E+04**	−1.35E+04	−1.08E+04	−1.22E+04	−1.48E+04
	Std	**2.83E+02**	8.11E+02	9.92E+02	1.06E+03	4.19E+02
TF9	Ave	**0.00E+00**	0.00E+00	7.84E+01	7.88E+01	1.02E+02
	Std	**0.00E+00**	0.00E+00	1.64E+01	2.52E+01	1.49E+01
TF10	Ave	**4.44E−15**	9.69E−12	1.20E+00	3.48E+00	1.90E−01
	Std	**0.00E+00**	6.13E−12	7.29E−01	8.28E−01	4.48E−01
TF11	Ave	**0.00E+00**	0.00E+00	1.28E−02	9.05E−02	2.70E−03
	Std	**0.00E+00**	0.00E+00	1.30E−02	4.07E−02	5.10E−03
TF12	Ave	**3.27E−03**	8.50E−03	3.19E−02	8.54E+00	1.87E−02
	Std	**2.77E−03**	5.20E−03	5.60E−02	7.76E−02	6.35E−02
TF13	Ave	**2.81E−01**	4.90E−01	4.19E−01	5.99E+01	1.83E−03
	Std	**1.38E−01**	1.93E−01	5.81E−01	1.67E+01	4.10E−03

MPA [39]: Step scaling factor = 0.5.
PSO [42]: Cognitive constant = 2, and social constant = 2.
SSA [43]: Leader position update probability = 0.5.
SHADE [44]: $P_{best} = 0.1$, and $rate_{Arc} = 2$.

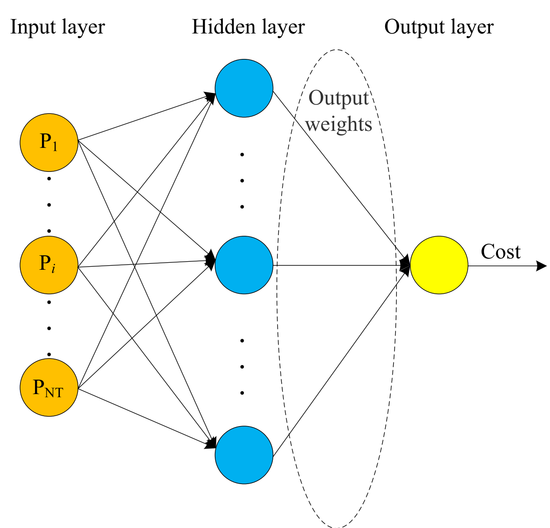

Fig. 4. A typical construction of ELM surrogate model.

proposed for solving combined heat and power ED problem. This improved MPA obtains more economical dispatching decisions. However, compared with this random possibility-based MPA, Q-MPA can handle the dispatching problems with uncertain wind energy. Furthermore, Q-MPA can be fit to dispatching problems with variational constraints, i.e., more strong robustness and scalability. Thus, Q-MPA is more suitable to solve CEUCPs while obtaining excellent performance.

3.3. ELM surrogate model to replace cost function

In the training of AC, much time is consumed by the evaluation of reward value or fitness value. Even offline actor network has been trained well, the online decision-making process is also need to be executed quickly. The reason is that real-time intra-day scheduling often occurs when an emergency arises. More seriously, the addition of uncertain wind energy further increases scheduling time. Thus, saving decision-making time is an indispensable link in CEUCPs. However, previous studies often save operating time by sacrificing accuracy [45]. This is undoubtedly a huge economic waste. In conclusion, a fast and accurate method to solve economic dispatch problem in CEUCP is necessary. To take up this challenge, time-consuming cost function is replaced by a time-saving ELM surrogate model.

(a) The construction of ELM-based surrogate models

As shown in Fig. 4, ELM is a kind of single hidden layer feedforward neural network (SHLFNN). ELM obtains input layer weights and hidden layer bias randomly. This is differ from original SHLFNN training method. The output layer weights are calculated analytically by minimizing the loss function. The loss function is composed by training error and output layer weight norm regular term. The analytic calculation relies on Moore–Penrose generalized inverse matrix, it is calculated analytically. Compared with other artificial neural networks, it obtains better generalization performance. In CEUCPs, fuel cost functions in different time period are quite different. Thus, ELM is suitable to replace the fuel cost function in CEUCPs. Assuming that NT units participate in scheduling, the training data are the vectors of the historical outputs in NT units. The output is set as 0 when a unit is closed. The label is the total cost of NT units. Training is described as 5 steps: ① input value multiplied by weight value, ② plus bias value, ③ compute activation function, ④ calculated output value, ⑤ assign the weight of output layer by matrix inversion.

(b) Modify ELM surrogate models by shape distance

In ELM surrogate model, the parameters of input layer are generated randomly. Thus, the accuracy may not be high enough. To overcome this drawback, shape distance is regarded as an additional convergence condition. The brief introduction of a simplified shape distance in the proposed ELM surrogate model is shown as following equations.

$$\begin{cases} S_{out} = \{s_1, s_2, \ldots, s_m\} \\ D = \{d_1, d_2, \ldots, d_j, \ldots, d_{m-1}\} \\ d_{j-1} = 0, \text{if } s_{out,j} - s_{out,j-1} = 0 \\ d_{j-1} = 1, \text{if } 0.5 > s_{out,j} - s_{out,j-1} > 0 \\ d_{j-1} = 2, \text{if } 1 > s_{out,j} - s_{out,j-1} > 0.5 \\ d_{j-1} = -1, \text{if } -0.5 < s_{out,j} - s_{out,j-1} < 0 \\ d_{j-1} = -2, \text{if } -1 < s_{out,j} - s_{out,j-1} < -0.5 \end{cases} \tag{22}$$

where S_{out} is a normalized sequence, $S_{out,j}$ is jth element of S_{out}, m represents total m elements, D is shape sequence and d_j is jth element of D. The original ELM algorithm is revised and shown as Algorithm 4.

Algorithm 4: Shape distance-assisted ELM model

Input: N_{input} sequences of NT units
Output: Parameters of ELM surrogate model
Calculate true cost values (Cost_t) of input sequences
Compute shape distances ($S_{\text{out,t}}$) of normalized Cost_t
while *Different elements not less than* $N_{input}/10$ **do**
    Execute original ELM algorithm
    Calculate cost values (Cost_v) by the ELM model
    Normalize the obtained Cost_v
    Calculate corresponding shape distances ($S_{\text{out,v}}$)
    Count different elements between $S_{\text{out,v}}$ and $S_{\text{out,t}}$
end

3.4. Handling methods of constraints in CEUCP

Handling methods of constraints in CEUCP can be divided into two portions, i.e., handling methods in status of units (discrete variable constraint) and handling methods in outputs of units (continuous variable constraint). For discrete variable constraints, i.e., the minimum start/stop hours and startup cost constraints, are handled in AC reinforcement learning framework.

(a) For the minimum start hours constraint, for an online unit i at time period $(t-1)$, it can still be online at time period (t). The unit state $I_i(t)$ is equal to 1 ($I_i(t-1)$ is equal to 1). If this online unit i is transferred to offline state at time period (t), the online time $h_{\mathrm{on},i}(t)$ must be equal to or greater than the minimum online time of $T_{\mathrm{ON},i}$. Then, the unit state $I_i(t)$ is equal to 0 ($I_i(t-1)$ is equal to 1).

(b) For the minimum stop hours constraint, for an offline unit i at time period $(t-1)$, it can still be offline at time period (t). The unit state $I_i(t)$ is equal to 0 ($I_i(t-1)$ is equal to 0). If this offline unit i is transferred to online state at time period (t), the offline time $h_{\mathrm{OFF},i}(t)$ must be equal to or greater than the minimum offline time of $T_{\mathrm{OFF},i}$. Then, the unit state $I_i(t)$ is equal to 1 ($I_i(t-1)$ is equal to 0).

(c) The hot and cold startup cost constraints are handled simultaneously when handling the minimum stop hours constraint. As shown in (8), if the offline time $h_{\mathrm{OFF},i}(t)$ is greater than the sum of the minimum offline time of $T_{\mathrm{OFF},i}$ and cold start time period CH_i, the ith unit presents a cold start state. Then, the total cost of CEUCP is increased by CC_i. If the offline time $h_{\mathrm{OFF},i}(t)$ is equal to or less than the sum of the minimum offline time of $T_{\mathrm{OFF},i}$ and cold start time period CH_i, the ith unit presents a hot start state. Then, the total cost of CEUCP is increased by HC_i.

For continuous variable constraints, i.e., power balance limit, generation limit, spinning reserve constraints, ramping up/down capacity constraints, and power output limits on wind energy system, are handled in the calculation of reward in AC reinforcement learning framework. Due to some constraints cannot be satisfied with inappropriate unit states, the constraint processing involves modifying discrete variables.

(d) For handling generation limit and ramping up/down capacity constraints, (4) and (6) can be integrated into:

$$\begin{cases} \max\{P_i^{\min}(t)\times I_i(t), P_i^s(t-1)-DR_i^s(t)\} \le P_i^s(t) \\ P_i^s(t) \le \min\{P_i^{\max}(t)\times I_i(t), P_i^s(t-1)+UR_i^s(t)\} \end{cases} \tag{23}$$

Then, if $P_i^s(t)$ exceeds the upper bound, $P_i^s(t)$ is assigned upper limit. If $P_i^s(t)$ is less than the lower bound, $P_i^s(t)$ is assigned lower bound.

(e) For handling the spinning reserve constraints, the on/off states of units may change. If the spinning reserve capacity is not enough, a certain unit that is turned off needs to be turned on. Then, the unit state should be redetected and processed by the aforementioned (a), (b), and (c). Turn on the units one by one as described above until the spinning reserve capacity meets the requirement.

(f) For solving the power balance limit and power output limits on wind energy system, an equality constraint processing method based on loop iteration is employed. Firstly, the sum value $P_{\mathrm{sum}}^s(t)$ of NT units ($P_1^s(t)$, $P_2^s(t)$,…, $P_{\mathrm{NT}}^s(t)$) are obtained by simple addition. Secondly, the error $E_{\mathrm{sum}}^s(t)$ is obtained by:

$$E_{\mathrm{sum}}^s(t) = P_{\mathrm{sum}}^s(t) - P_{\mathrm{load}}(t) - P_{\mathrm{loss}}^s(t) + W^s(t) \tag{24}$$

Thirdly, $E_{\mathrm{sum}}^s(t)$ is divided into N_{su} parts and assigned to N_{su} unit, where N_{su} is the number of units with surplus capacity. Finally, steps 1 through 3 are looped until the error $E_{\mathrm{sum}}^s(t)$ is less than the preset threshold. It should be noted that the unit state may be changed. If all the units take the lower limit of capacity and $E_{\mathrm{sum}}^s(t)$ is less than 0, a certain online unit should be transferred to the offline state. If all the units take the upper limit of capacity and $E_{\mathrm{sum}}^s(t)$ is greater than 0, a certain offline unit should be transferred to the online state. Then, the unit state should be redetected and processed by the aforementioned (a), (b), and (c).

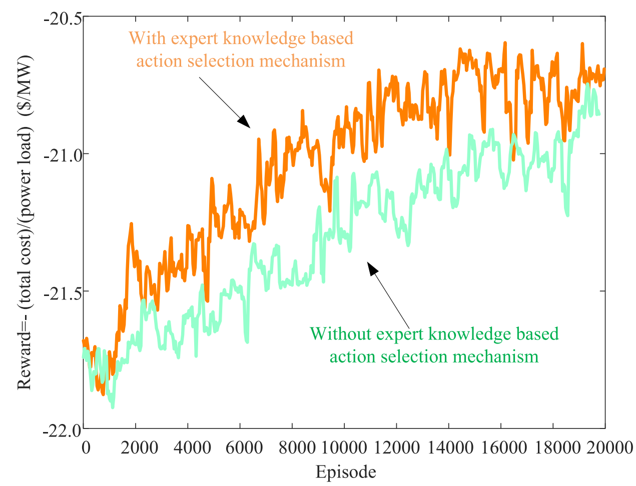

Fig. 5. Convergence curves of AC training in case 1.

4. Simulation and analysis

In this section, effectiveness and robustness of the proposed method is demonstrated through simulations of 10-unit and 100-unit test system with uncertain wind energy. Case 1 involves a system containing ten thermal units, with data sourced from [46]. Case 2 comprises 100 units, obtained by replicating case 1 ten times. In case 3, the data for uncertain wind energy is derived from [33]. The wind turbines are scaled up tenfold to correspond with the replication of case 1. The numbers of population and iteration of Q-MAP in all three cases are set to 100. Parameters for the AC and ELM surrogate models remain consistent with the descriptions provided in Section 3.

4.1. Case 1: Results of IEEE 10 units test system

This case has been widely studied, and it is employed to compare with other researches for illustrating the superiority of this scheme. The power demands for a 24 hour period are sourced from [46]. The construction of the critic network in expert knowledge data-driven based AC is shown as follows:

• The size of input layer is 20;
• The size of convolution kernel is 3 × 1;
• A rectified linear unit (ReLU) layer;
• The length of pooling window in pooling layer is 2 × 1;
• The nodes of the 1st fully connected layer (FCL) is 40;
• The nodes of the 2nd FCL is 128;
• The nodes of the 3rd FCL is 64;
• The node of the regression layer is only 1 (the Q value).

The construction of actor network is shown as follows:

• The size of input layer is 10;
• The size of convolution kernel is 3 × 1;
• A rectified linear unit (ReLU) layer;
• The length of pooling window in pooling layer is 2 × 1;
• The nodes of the 1st FCL are 40;
• The nodes of the 2nd FCL are 128;
• The nodes of the 3rd FCL are 64;
• The nodes of the regression layer are 10 (predicted action of up/down).

The construction of ELM-based surrogate model is shown as follows:

• Nodes of input layer are 10;
• Nodes of middle layer (hidden layer) are 35;
• Node of output layer is 1 (fitted cost value);

The optimal on/off status and dispatching decisions during 24 h are presented in Table 2. Further, in Table 2, A_1 and A_2 represent the accuracy of improved ELM and original ELM surrogate models at

Table 2
Best solutions (hourly generation: MW) and the accuracies of surrogate models in case 1.

Hour	1	2	3	4	5	6	7	8	9	10	11	12
Unit 1	455.00	455.00	455.00	455.00	455.00	455.00	455.00	455.00	455.00	455.00	455.00	455.00
Unit 2	245.00	295.00	370.00	443.92	435.05	360.00	410.00	455.00	455.00	455.00	455.00	455.00
Unit 3	0	0	0	0	0	130.00	130.00	130.00	130.00	130.00	130.00	130.00
Unit 4	0	0	0	0	84.95	130.00	130.00	130.00	130.00	130.00	130.00	130.00
Unit 5	0	0	25.00	51.08	25.00	25.00	25.00	30.00	85.00	162.00	162.00	162.00
Unit 6	0	0	0	0	0	0	0	0	20.00	33.00	73.00	80.00
Unit 7	0	0	0	0	0	0	0	0	25.00	25.00	25.00	25.20
Unit 8	0	0	0	0	0	0	0	0	0	10.00	10.00	42.77
Unit 9	0	0	0	0	0	0	0	0	0	0	10.00	10.04
Unit 10	0	0	0	0	0	0	0	0	0	0	0	10.00
A_1 (%)	94.74	94.94	95.16	95.11	95.42	94.96	94.91	95.03	94.75	95.30	94.97	95.52
A_2 (%)	91.96	94.57	92.02	93.49	93.52	92.81	92.01	91.69	93.16	93.48	92.19	93.10
Hour	13	14	15	16	17	18	19	20	21	22	23	24
Unit 1	455.00	455.00	455.00	455.00	455.00	455.00	455.00	455.00	455.00	455.00	455.00	455.00
Unit 2	455.00	455.00	455.00	310.00	260.00	360.00	455.00	455.00	455.00	455.00	399.95	345.00
Unit 3	130.00	130.00	130.00	130.00	130.00	130.00	130.00	130.00	130.00	0	0	0
Unit 4	130.00	130.00	130.00	130.00	130.00	130.00	130.00	130.00	130.00	0	0	0
Unit 5	162.00	85.00	30.00	25.00	25.00	25.00	30.00	162.00	85.00	145.00	45.05	0
Unit 6	33.00	20.00	0	0	0	0	0	33.00	20.00	20.00	0	0
Unit 7	25.00	25.00	0	0	0	0	0	25.00	25.00	25.00	0	0
Unit 8	10.00	0	0	0	0	0	0	10.00	0	0	0	0
Unit 9	0	0	0	0	0	0	0	0	0	0	0	0
Unit 10	0	0	0	0	0	0	0	0	0	0	0	0
A_1 (%)	94.69	95.56	95.12	94.99	94.78	95.00	95.00	95.22	95.34	94.87	95.17	94.78
A_2 (%)	92.68	92.18	93.88	92.22	92.04	92.73	94.05	92.57	93.66	92.43	91.77	94.56

Table 3
Comparison of Q-MPA with other approaches for case 1.

Approach	Best ($)	Average ($)	Worst ($)	Std. (%)	Avg. Time (Second)
EP [47]	564,511	565 352	566,231	0.3	100
SA [48]	565,828	565,998	566,260	0.08	3
GACC [49]	563,977	564,791.5	565,606	0.0857	85
QEA [50]	563,938	563,969	564,672	0.13	19
hGADE [51]	563,938	564,044	564,283	0.06	24
BGWO [49]	563,976.6	564,378.58	565,518.1	0.106	64.19
Q-MPA	**562,980.06**	**563,021.30**	**563,088.13**	**0.02**	**0.42**

Table 4
Evaluation time of ELM model and original function in case 1.

Model	Average time (s)
Original cost function	3.48E−05
ELM surrogate model	2.90E−07

each time period, respectively. The shadow indicates that the unit is in operation. The number 0 represents that the unit is not in operation. The application of shape distance improves accuracy by nearly 2% when compared to the original ELM. Thus, the usage of surrogate model is time-saving, and the shape distance is an effective convergence condition. Optimal cost and average execution time are 562,980.06 $ and 0.42 s, respectively. Comparative results are displayed in Table 3. Compared with EP (evolutionary programming) [47], SA (simulated annealing) [48], GACC (genetic algorithm based on characteristic classification) [49], QEA (quantum-inspired evolutionary algorithm) [50], hGADE (a hybrid of genetic algorithm and differential evolution) [51], and BGWO (binary grey wolf optimizer) [49], the best total cost is reduced by 0.27%, 0.50%, 0.18%, 0.17%, 0.17%, and 0.18%, respectively. The results demonstrate the excellent convergence of the proposed algorithm.

Moreover, the on/off status can be rapidly acquired using a convolutional neural network (actor network in AC). The convergence curves of the AC, both with and without the expert knowledge based action selection mechanism, are depicted as Fig. 5. The convergence speed employing the proposed action selection mechanism surpasses that of the AC without the proposed mechanism. Consequently, examining additional optional actions promotes accelerated convergence of AC methods.

Then, let us focus on the execution time. In comparison to other algorithms, average execution time is reduced by 99.58%, 80.60%, 99.51%, 97.79%, 98.25%, and 99.35%, respectively. Compared to the original, time-consuming computation of cost functions, a significant amount of execution time is saved by employing surrogate models. Numerically, the execution time of original cost function and improved ELM is shown as Table 4, revealing a 99.17% reduction in time. In conclusion, the proposed AC method of solving CEUCPs is effective and time-saving.

4.2. Case 2: Results of IEEE 100 units test system

Large-scale units are involved in Case 2, and the original UCPs are transferred to CEUCPs. The power demand is multiplied by 10 times of case 1. Then, this UCP transfers to a CEUCP. The construction of the critic network in expert knowledge data-driven based AC is shown as follows:

- The size of input layer is 200;
- The size of convolution kernel is 3 × 1;
- A rectified linear unit (ReLU) layer;
- The size of pooling window in pooling layer is 2 × 1;
- The nodes of the 1st FCL is 60;
- The nodes of the 2nd FCL is 192;
- The nodes of the 3rd FCL is 96;
- The node of the regression layer is only 1 (the Q value).

The construction of actor network is shown as follows:

- The size of input layer is 100;
- The size of convolution kernel is 3 × 1;
- A rectified linear unit (ReLU) layer;

Table 5
Comparison of Q-MPA with other approaches for case 2.

Approach	Best ($)	Average ($)	Worst ($)	Std. (%)	Avg. Time (Second)
EP [47]	5,623,885	5,633,800	5,639,148	0.27	6120
SA [48]	5,617,876	5,624,301	5,628,506	0.19	696
GACC [49]	5,626,514	5,636,522	5,646,529	0.1012	3547
QEA [50]	5,609,550	5,611,797	5,613,220	0.07	80
hGADE [51]	5,604,787	5,610,074	5,620,236	0.27	397
BGWO [49]	5,628,975	5,637,659	5,643,899	0.08131	836.54
Q-MPA	**5,580,622.55**	**5,580,988.31**	**5,581,324.16**	**0.04**	**1.81**

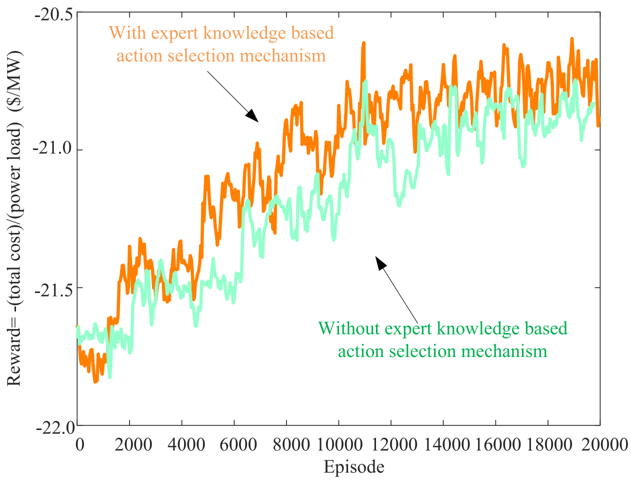

Fig. 6. Convergence curves of AC training in case 2.

Table 6
Evaluation time of ELM model and original function in case 2.

Model	Average time (s)
Original cost function	5.97E−05
ELM surrogate model	3.37E−06

Table 7
Accuracies of surrogate models in case 2.

Hour	1	2	3	4	5	6
A_1 (%)	94.69	95.56	95.12	94.99	94.78	95.00
A_2 (%)	92.68	92.18	93.88	92.22	92.04	92.73
Hour	7	8	9	10	11	12
A_1 (%)	94.69	95.56	95.12	94.99	94.78	95.00
A_2 (%)	92.68	92.18	93.88	92.22	92.04	92.73
Hour	13	14	15	16	17	18
A_1 (%)	94.69	95.56	95.12	94.99	94.78	95.00
A_2 (%)	92.68	92.18	93.88	92.22	92.04	92.73
Hour	19	20	21	22	23	24
A_1 (%)	95.00	95.22	95.34	94.87	95.17	94.78
A_2 (%)	94.05	92.57	93.66	92.43	91.77	94.56

- The size of pooling window in pooling layer is 2 × 1;
- The nodes of the 1st FCL is 60;
- The nodes of the 2nd FCL is 192;
- The nodes of the 3rd FCL is 96;
- The nodes of the regression layer are 100 (predicted action of up/down).

The construction of ELM-based surrogate model is shown as follows:

- Nodes of input layer are 100;
- Nodes of middle layer (hidden layer) are 155;
- Node of output layer is 1 (fitted cost value);

The best cost and average execution time are only 5,580,622.55 $ and 1.81 s, respectively. Firstly, as shown in Table 5, compared with EP, SA, GACC, QEA, hGADE, and BGWO, the best total cost is reduced by 0.77%, 0.66%, 0.82%, 0.52%, 0.43%, and 0.86%, respectively. Thus, the convergence of this method is excellent for large-scale power system. In other words, the performance improvement of the proposed method is more pronounced for CEUCPs. Thanks to the off-line AC networks, the on/off status can be obtained quickly by a CNN. Further, the convergence curves of AC with or without the expert knowledge based action selection mechanism are shown as Fig. 6. The proposed expert knowledge based action selection mechanism is also effective in AC for CEUCPs.

Compared with other algorithms, average execution time is reduced by 99.97%, 99.74%, 99.95%, 97.74%, 99.54%, and 99.78%, respectively. Further, the execution time of original cost function and improved ELM is shown as Table 6. The evaluation time is reduced by 94.35%. Further, the accuracy of improve ELM and original ELM is shown as Table 7. Compared with original ELM, the application of shape distance improves the accuracy by nearly 3%. Thus, the usage of surrogate model is time-saving, and the shape distance is an effective convergence condition. In summary, the proposed method of solving CEUCPs is effective and time-saving.

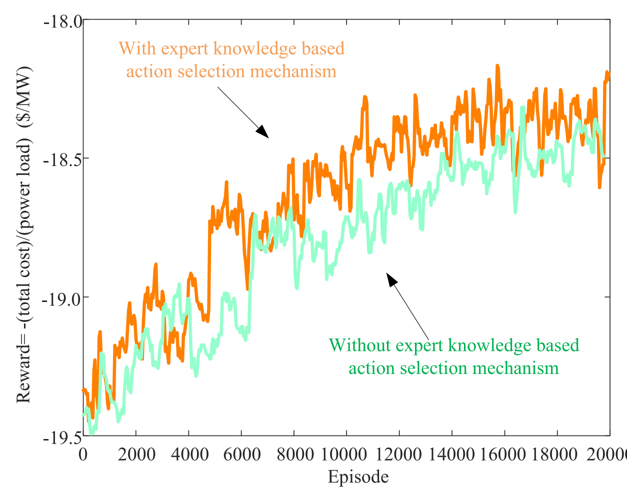

Fig. 7. Convergence curves of AC training in case 3.

4.3. Case 3: Results of IEEE 100 units test system with uncertain wind energy

Uncertain wind energy is added to case 2 in this case, which is used to explain the effectiveness of the proposed scheme for CEUCPs with renewable energy. The power demands are the same as case 2. Twenty scenarios of uncertain wind energy are generated in Fig. 2(a). The construction of the critic network in AC is shown as follows:

- The size of input layer is 200;
- The size of convolution kernel is 3 × 1;
- A rectified linear unit (ReLU) layer;
- The size of pooling window in pooling layer is 2 × 1;
- The nodes of the 1-st FCL is 80;

Table 8
Statistical results of Q-MPA for case 3 (time: second).

Best ($)	Average ($)	Worst ($)	Std. (%)	Avg. Time
4,949,588.14	4,949,671.88	4,949,803.47	0.04	1.96

Table 9
Accuracies of surrogate models in case 3.

Hour	1	2	3	4	5	6
A_1 (%)	94.69	95.56	95.12	94.99	94.78	95.00
A_2 (%)	92.68	92.18	93.88	92.22	92.04	92.73
Hour	7	8	9	10	11	12
A_1 (%)	95.00	95.22	95.34	94.87	95.17	94.78
A_2 (%)	94.05	92.57	93.66	92.43	91.77	94.56
Hour	13	14	15	16	17	18
A_1 (%)	94.69	95.56	95.12	94.99	94.78	95.00
A_2 (%)	92.68	92.18	93.88	92.22	92.04	92.73
Hour	19	20	21	22	23	24
A_1 (%)	95.00	95.22	95.34	94.87	95.17	94.78
A_2 (%)	94.05	92.57	93.66	92.43	91.77	94.56

Table 10
Evaluation time of ELM model and original function in case 3.

Model	Average time (s)
Original cost function	7.44E−05
ELM surrogate model	3.52E−06

- The nodes of the 2-nd FCL is 256;
- The nodes of the 3-rd FCL is 128;
- The node of the regression layer is only 1 (the Q value).

The construction of actor network is shown as follows:

- The size of input layer is 100;
- The size of convolution kernel is 3 × 1;
- A rectified linear unit (ReLU) layer;
- The size of pooling window in pooling layer is 2 × 1;
- The nodes of the 1-st FCL is 80;
- The nodes of the 2-nd FCL is 256;
- The nodes of the 3-rd FCL is 128;
- The nodes of the regression layer are 100.

The construction of ELM-based surrogate model is the same as the setting of case 2. The best cost and execution time shown in Table 8 are only 4,949,588.14 $ and 1.96 s, respectively. Further, the convergence curves of AC with or without action selection mechanism are shown as Fig. 7. The convergence speed is higher than AC without the proposed action selection mechanism. Thus, testing more optional actions contributes to fast convergence in AC for CEUCPs with uncertain wind energy. Numerically, as shown in Table 10, the execution time of original cost function and improved ELM surrogate model are 7.44E−05 and 3.52E−06 s, respectively. The execution time is reduced by 95.27%. As shown in Table 9, the usage of shape distance improves accuracy by nearly 3%. Thus, the ELM model is also time-saving, effective and accurate. In conclusion, this method can obtain feasible solutions in CEUCPs within several seconds.

Compared with case 2, total cost is reduced by 11.30%. It highlights the economic advantages of integrated-wind energy. The cost-effectiveness of wind energy implies that greater emphasis will be placed on incorporating renewable energy sources within the power grids. As the global emphasis on sustainable energy solutions and carbon emission reduction intensifies, the proportion of renewable energy is anticipated to increase significantly in future power grid systems. Consequently, this will lead to the development of more sustainable, green, and economic power networks, thereby contributing to global efforts in combating climate change and promoting energy security.

4.4. Performances of the proposed method

To solve the CEUCPs with uncertain wind energy, an expert knowledge data-driven based AC reinforcement learning method is proposed in this paper. The effectiveness, superiority, and scalability are discussed in this subsection.

(a) *Effectiveness of the proposed method*

The simulations of 10-units, 100-units, and 100-units with wind energy test systems are employed to verify effectiveness of the proposed method. In case 1, 2, and 3, firstly, after 2×10^4 steps of training, the value of reward reaches the convergence state. By the strong inference of the proposed expert knowledge data-driven based AC reinforcement learning method, the unit states can be obtained without constraint violation. In other words, the feasible on/off states are obtained for safe operation in power systems. It is worth noting that for the large-scale system with or without wind energy uncertainty, the proposed method can also obtain feasible on/off states. The adaptability of large-scale power grid makes it worth applying the proposed method to real-world engineering projects. Thus, the proposed expert knowledge is an effective innovation in AC reinforcement learning method. Secondly, the outputs of units are obtained precisely by the proposed Q-MPA. Thanks to the quantum-based search pattern, the Q-MPA can execute CEUCPs effectively. Finally, the execution time of obtaining final decisions is short with the usage of ELM data-driven surrogate model. The short operation time fully alleviates the computing pressure caused by dispatching cycle in a power system. To sum up, the proposed AC reinforcement learning method is effective to solve CEUCPs.

(b) *Scalability of the proposed method*

To illustrate the scalability, except for testing in 10-units system of case 1, the proposed method is tested in 100-units without or with wind energy test systems. In case 2, compared with case 1, the scale of units is expanded by 10 times. Due to the unaffordable number of integer combination, large-scale test system is difficult to obtain the optimal unit on/off states. More variables are included which makes optimization of CEUCPs difficult to converge. Thanks to the strong inference ability, the expert knowledge based AC reinforcement learning framework is able to obtain feasible and economical unit on/off states. Then, continuous variables in CEUCPs (generating outputs of units) are also difficult to be obtained. The powerful search ability of Q-MPA is shown in this high-dimension optimization problem. Due to the strong convergence of Q-MPA, except for small-scale 10-units test system, more stable and economical dispatching decisions are obtained in the computationally expensive optimization of large-scale 100-units test system. Finally, the CEUCP in 100-units test system exist computationally expensive characteristic. This will result in time-consuming optimization execution which may exceed 5 or 15 min dispatching cycle. Thanks to the usage of shape distance-based ELM surrogate model, the dispatching time is reduced to a great extent. The ELM surrogate model can not only realize the fitting of objective function in low-dimension optimization problem, but also the fitting of large-scale, high-dimension, and computationally expensive objective function in CEUCP. To sum up, considering a large-scale test system with more units, scalability of the proposed method is strong.

Furthermore, in case 3, the random wind energy can only be predicted in real time. Thus, three requirements of scalability emerge [52], i.e., ① powerful real-time inferential capability for obtaining the optimal unit on/off states, ② strong convergence of Q-MAP to realize the optimization with different wind energy, and ③ strong fitting and scalable ability of ELM surrogate model across different time period for solving CEUCP. Fortunately, the proposed method in this paper meets the above three requirements of scalability. Firstly, the proposed expert knowledge based AC reinforcement learning framework is able to determine unit on/off states with different wind energy scenario at different time period. Secondly, the Q-MPA obtains excellent convergence. Thus, in case 3, more economical dispatching decisions can be obtained with time-varying constraints (different outputs of wind farms). Finally, proposed ELM surrogate model is trained offline. As shown in Table 9, even taking into account time-varying wind energy, the accuracies of surrogate models across different time periods are high. In other words, the strong fitting and scalable ability of ELM surrogate model exist across different time period for solving CEUCP. To sum up, scalability of proposed method is excellent.

(c) *Superiority of the proposed method*

The superiority of the proposed method can be divided by three aspects: ① the superiority of the proposed expert knowledge data-driven based actor–critic reinforcement learning framework, ② the superiority of Q-MPA, and ③ the superiority of employing ELM surrogate model.

Firstly, the proposed deep RL method is time-saving to solve CEUCPs. Compared with meta-heuristic algorithms, i.e., EP [47], SA [48], GACC [49], QEA [50], hGADE [51], and BGWO (binary grey wolf optimizer) [49], the proposed deep RL method obtains the results of CEUCPs meta-heuristic algorithm in the shortest time. The reason is that the optimization time of determining the units on/off status is transferred to the offline RL models training time. Thus, the proposed expert knowledge data-driven based actor–critic reinforcement learning framework is time-saving. Furthermore, the expert knowledge based action selection mechanism is effective to enhance the training speed and model performance of proposed RL method. In case 1, 2, and 3, compared with not using the expert knowledge based action selection mechanism (traditional AC [18]), the proposed AC method obtains better rate and degree of convergence. Thanks to the usage of expert knowledge, expert knowledge with the heuristic of similar generating unit is effective to enhance the performance of actor–critic reinforcement learning framework for solving CEUCPs.

Secondly, the proposed Q-MPA is effective. In the TF1-TF13 testing functions, compared with MPA [39], PSO [42], SSA [43], and SHADE [44], the proposed Q-MPA obtains the best convergence and robustness. The convergence is associated with the economy of dispatching decisions in CEUCPs. The robustness is associated with the ability of obtain dispatching decisions in CEUCPs stably. Thus, compared with other meta-heuristic algorithms, the proposed Q-MPA is more suitable for solving CEUCPs. Then, the proposed Q-MPA obtains the best economical dispatching decisions of CEUCPs in all case 1, 2, and 3. Thanks to the excellent convergence of Q-MPA, less generating cost can be obtained by Q-MPA. To sum up, the proposed Q-MPA is effective to solve CEUCPs.

Finally, compared with applying original mathematical objective function, employing ELM surrogate model is more time-saving. By using ELM surrogate models, more than 94% evaluation time of objective functions can be reduced in case 1, 2, and 3. More important, the accuracy of proposed surrogate model can be enhanced by integrating shape distance as an additional convergence condition. Different from sacrificing accuracy for reducing operating time [45], the proposed ELM surrogate models are both time-saving and accurate. To sup up, by using the proposed ELM surrogate models, CEUCPs can be solved quickly and accurately.

5. Concluding remarks

This paper proposes an expert knowledge data-driven based AC reinforcement learning method to solve CEUCPs with uncertain wind energy. The original CEUCP is converted into a MDP process. Then, an AC framework which is trained by historical operational data is proposed to determine unit status. Due to the strong ability of AC to learn the characteristics in sequences, the obtained status of units are accurate, economic, and robust. To reduce training time, a novel expert knowledge based action selection mechanism is integrated into AC. The expert knowledge based action selection mechanism speeds up the training process and enhances convergence of AC. The status of units are obtained by the well trained AC framework. Execution time of CEUCPs with uncertain wind energy is reduced. Then, the dispatching decisions are computed by a Q-MPA. Compared with original MPA, the convergence and robustness are enhanced by quantum based search pattern. The improved algorithm is more suitable for solving CEUCPs. More importantly, to further reduced the training and execution time, an improved ELM surrogate model is proposed. The evaluation time of ELM surrogate model is less than the evaluation time of numerical function. The advantage of this method is verified by 10-units and 100-units with or without uncertain wind energy test systems. The results show that proposed expert knowledge data-driven based AC reinforcement learning method is effective to solve CEUCPs with wind energy. Based on the following findings, it remains a challenge to uncover data-driven RL methods for obtaining the units status and generating outputs simultaneously. Furthermore, for environmental purposes, other renewable energies, storage batteries, and hydrogen production can be integrated into CEUCPs, and these problems are left for future work.

CRediT authorship contribution statement

Huijun Liang: Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization. **Chenhao Lin:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation. **Aokang Pang:** Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijepes.2024.110033.

References

[1] Zare M, Malekpour M, Azizipanah-Abarghooee R, Terzija V. Stochastic unit commitment to determine frequency response ramp rate including wind turbines with synthetic inertia and virtual synchronous generator. Int J Electr Power Energy Syst 2023;152:109272.
[2] Roy P. Solution of unit commitment problem using gravitational search algorithm. Int J Electr Power Energy Syst 2013;53:85–94.
[3] Senjyu T, Shimabukuro K, Uezato K, Funabashi T. A fast technique for unit commitment problem by extended priority list. IEEE Trans Power Syst 2003;18(2):882–8.
[4] Ostrowski J, Anjos MF, Vannelli A. Tight mixed integer linear programming formulations for the unit commitment problem. IEEE Trans Power Syst 2012;27(1):39–46.
[5] Roy P, Sarka R. Solution of unit commitment problem using quasi-oppositional teaching learning based algorithm. Int J Electr Power Energy Syst 2014;60:96–106.
[6] Yuan X, Ji B, Zhang S, Tian H, Hou Y. A new approach for unit commitment problem via binary gravitational search algorithm. Appl Soft Comput 2014;22:249–60.
[7] Yang Y, Li G, Luo T, Al-Bahrani M, Al-Ammar E, Sillanpaa M, et al. The innovative optimization techniques for forecasting the energy consumption of buildings using the shuffled frog leaping algorithm and different neural networks. Energy 2023;268:126548.
[8] Ruan G, Wang J, Zhong H, Xia Q, Kang C. Improving sample efficiency of deep learning models in electricity market. IEEE Trans Power Syst 2022;38(5):4761–73.
[9] Li M, Wang K, He S. Maximizing energy efficiency by optimizing relay deployment in EH-WSNs for smart grid. IEEE Commom Lett 2023;27(2):625–9.
[10] Wang Y, Hu J, Liu N. Energy management in integrated energy system using energy-carbon integrated pricing method. IEEE Trans Sustain Energy 2023;14(4):1992–2005.

[11] Shao Z, Zhai Q, Han Z, Guan X. A linear AC unit commitment formulation: An application of data-driven linear power flow model. Int J Electr Power Energy Syst 2023;145:108673.
[12] Bhavsar S, Pitchumani R, Ortega-Vazquez M, Costilla-Enriquez N. A hybrid data-driven and model-based approach for computationally efficient stochastic unit commitment and economic dispatch under wind and solar uncertainty. Int J Electr Power Energy Syst 2023;151:109144.
[13] Hou W, Hou L, Zhao S, Liu W. A hybrid data-driven robust optimization approach for unit commitment considering volatile wind power. Electr Power Syst Res 2022;205:107758.
[14] Zhou Y, Zhai Q, Wu L, Shahidehpour M. A data-driven variable reduction approach for transmission-constrained unit commitment of large-scale systems. J Mod Power Syst Clean Energy 2023;11(1):254–66.
[15] Zhou X, Xu X, Liang W, Zeng Z, Yan Z. Deep-learning-enhanced multitarget detection for end–edge–cloud surveillance in smart IoT. IEEE Internet of Things J 2021;8(16):12588–96.
[16] Salman D, Kusaf M. Short-term unit commitment by using machine learning to cover the uncertainty of wind power forecasting. Sustainability 2021;13:13609.
[17] Mohammadi F, Sahraei M, Trakas D, Nikos D. Machine learning assisted stochastic unit commitment during hurricanes with predictable line outages. IEEE Trans Power Syst 2021;36(6):5131–42.
[18] Mars P, Sullivan A. Applying reinforcement learning and tree search to the unit commitment problem. Appl Energy 2021;302:117519.
[19] Ajagekar A, You F. Deep reinforcement learning based unit commitment scheduling under load and wind power uncertainty. IEEE Trans Sustain Energy 2023;14(2):803–12.
[20] Patrick D, Aidan O. Reinforcement learning and A* search for the unit commitment problem. Energy AI 2022;9:100179.
[21] Gonzalez-Garci A, Barragan-Alcantar D, Collado-Gonzalez I, Garrido L. Adaptive dynamic programming and deep reinforcement learning for the control of an unmanned surface vehicle: Experimental results. Control Eng Pract 2021;111:104807.
[22] Zhou Z, Wu Z, Jin T. Deep reinforcement learning framework for resilience enhancement of distribution systems under extreme weather events. Int J Electr Power Energy Syst 2021;128:106676.
[23] Boute R, Gijsbrechts J, Jaarsveld W, Vanvuchelen N. Deep reinforcement learning for inventory control: A roadmap. European J Oper Res 2022;298(2):401–12.
[24] Tariq U, Ali H, Hussain M, Liu L. Shuffled ARSH-fati: A novel meta-heuristic for lifetime maximization of range-adjustable wireless sensor networks. IEEE Trans Green Commun Netw 2023;7(3):1217–33.
[25] Pan J, Hu P, Chu S. Binary fish migration optimization for solving unit commitment. Energy 2021;226:120329.
[26] Ponciroli R, Stauff N, Ramsey J, Ganda F, Vilim R. An improved genetic algorithm approach to the unit commitment/economic dispatch problem. IEEE Trans Power Syst 2020;35(5):4005–13.
[27] Ali E, Elazim S, Balobaid A. Implementation of coyote optimization algorithm for solving unit commitment problem in power systems. Energy 2023;263:125697.
[28] Huo Y, Li P, Ji H, Yu H, Yan J, Wu J, et al. Data-driven coordinated voltage control method of distribution networks with high DG penetration. IEEE Trans Power Syst 2023;38(2):1543–57.
[29] Chen Y, Lin G, Chen Z, Wen J, Granderson J. A simulation-based evaluation of fan coil unit fault effects. Energy Build 2022;263:112041.
[30] Pang A, Liang H, Lin C, Yao L. A surrogate-assisted adaptive bat algorithm for large-scale economic dispatch. Energies 2023;16(2):1011.
[31] Lin C, Liang H, Pang A. A fast data-driven optimization method of multi-area combined economic emission dispatch. Appl Energy 2023;337:120884.
[32] Su W, Ai Z, Liu J, Yang B, Wang F. Maintaining an acceptable indoor air quality of spaces by intentional natural ventilation or intermittent mechanical ventilation with minimum energy use. Appl Energy 2023;348:121504.
[33] Shukla A, Singh SN. Multi-objective unit commitment with renewable energy using hybrid approach. IET Renew Power Gener 2016;10(3):327–38.
[34] Li F, Qin J, Zheng W. Distributed Q-learning-based online optimization algorithm for unit commitment and dispatch in smart grid. IEEE Trans Cybern 2020;50(9):4146–56.
[35] Yuan Y, Dehghanpour K, Wang Z, Bu F. A joint distribution system state estimation framework via deep actor-critic learning method. IEEE Trans Power Syst 2023;38(1):796–806.
[36] Al-Betar M, Awadallah M, Makhadmeh S, Alyasseri Z, Al-Naymat G, Mirjalili S. Marine predators algorithm: A review, archives of computational methods in engineering. Arch Comput Methods Eng 2023;30:3405–35.
[37] Abdel-Basset M, El-Shahat D, Chakrabortty R, Ryan M. Parameter estimation of photovoltaic models using an improved marine predators algorithm. Energy Convers Manage 2021;227:113491.
[38] Das ARaiRDhalK, Ray S. An inclusive survey on marine predators algorithm: Variants and applications. Arch Comput Methods Eng 2023;30:3133–72.
[39] Faramarzi A, Heidarinejad M, Mirjalili S, Gandomi A. Marine predators algorithm: A nature-inspired metaheuristic. Expert Syst Appl 2020;152:113377.
[40] Zhao J, Li L, Yu J, Li Z. An improved marine predators algorithm for optimal reactive power dispatch with load and wind-solar power uncertainties. IEEE Access 2022;10:126664-73.
[41] Shaheen A, Elsayed A, Ginidi A, El-Sehiemy R, Alharthi M, Ghoneim S. A novel improved marine predators algorithm for combined heat and power economic dispatch problem. Alex Eng J 2022;61(3):1834–51.
[42] Wang X, Henshaw P, Ting D. Exergoeconomic analysis for a thermoelectric generator using mutation particle swarm optimization (M-PSO). Appl Energy 2021;294:116952.
[43] Baliarsingh S, Vipsita S, Muhammad K, Bakshi S. Analysis of high-dimensional biomedical data using an evolutionary multi-objective emperor penguin optimizer. Swarm Evol Comput 2019;48:262–73.
[44] Mohamed A, Hadi A, Jambi K. Novel mutation strategy for enhancing SHADE and LSHADE algorithms for global numerical optimization. Swarm Evol Comput 2019;50:100455.
[45] Wang S, Zhao C, Fan L, Bo R. Distributionally robust unit commitment with flexible generation resources considering renewable energy uncertainty. IEEE Trans Power Syst 2022;37(6):4179–90.
[46] Logenthiran T, Woo W, Phan V. Lagrangian relaxation hybrid with evolutionary algorithm for short-term generation scheduling. Int J Electr Power Energy Syst 2015;64:356–64.
[47] Juste K, Kita H, Tanaka E, Hasegawa J. An evolutionary programming solution to the unit commitment problem. IEEE Trans Power Syst 1999;14(4):1452–9.
[48] Simopoulos D, Kavatza S, Vournas C. Unit commitment by an enhanced simulated annealing algorithm. IEEE Trans Power Syst 2006;21(1):68–76.
[49] Panwar L, Reddy S, Verma A, Panigrahi B, Kumar R. Binary grey wolf optimizer for large scale unit commitment problem. Swarm Evol Comput 2018;38:251–66.
[50] Lau T, Chung C, Wong K, Chung T, Ho S. Quantum-inspired evolutionary algorithm approach for unit commitment. IEEE Trans Power Syst 2009;24(3):1503–12.
[51] Trivedi A, Srinivasan D, Biswas S, Reindl T. Hybridizing genetic algorithm with differential evolution for solving the unit commitment scheduling problem. Swarm Evol Comput 2015;23:50–64.
[52] Yi Z, Xu Y, Wu C. Model-free economic dispatch for virtual power plants: An adversarial safe reinforcement learning approach. IEEE Trans Power Syst 2024;39(2):3153–68.